

LegalEase: AI-Powered Legal Document Generator

Development & Integration

Project Phase	5. Project Development
Team	Team 3
Date	02 October 2026
Prepared From	Supplied LegalEase repository + supplied project templates

Module	Responsibility	Integration
frontend/app.py	Streamlit UI, validation, API call, preview, downloads	Calls FastAPI /generate and formatter functions
legalEaseAPI/routes.py	Request model and generation endpoint	Calls GeminiDocumentGenerator
ai_core/gemini_generator.py	Prompt construction and Gemini API	Returns document text
ai_core/generator.py	Sanitization and formatting	HTML preview, DOCX and PDF
config.py	Environment configuration	Model, API URL, key and logo path

Error handling

Client validation blocks incomplete forms. Connection failures are shown to the user. Generation failures become HTTP 502. The generator retries transient failures before raising GeminiGenerationError.

API key security

GEMINI_API_KEY is loaded from environment configuration. A real secret must never be committed.

Source basis: Implemented claims are based on the supplied LegalEase repository. Items without repository evidence are explicitly marked pending, future, or not implemented.